

CSMFAB Materials Study — Part I

Ultra-High Temperature Ceramics (UHTCs) and Advanced Structural Ceramics

Carrington Storm Motors | Safe Pod Engineering Company | June 2026

Study Scope: This document covers all ceramic-class materials used across the CSMFAB product portfolio. Each material is analyzed for: chemical composition, industrial synthesis, commercial acquisition, raw-material mining and harvesting, in-house synthesis feasibility, recycled-material pathways, and 2026 market pricing at multiple volume tiers.

Used In: CSMFAB products 001, 002, 003, 006, 007, 009, 010, 015, 016, 019, 020, 021, 031, CSMIC 105, 107, CSMMECH 001–003

Material 1 — Zirconium Diboride (ZrB₂)

Formula: ZrB₂ | **Crystal:** Hexagonal (AlB₂ type) | **Class:** UHTC

1.1 Physical and Chemical Properties

- Melting point: 3245°C (highest of common UHTCs)
- Density: 6.09 g/cm³
- Electrical resistivity: 7.8–9.2×10⁻⁸ Ω·m (mildly conductive — requires sub-percolation loading)
- Thermal conductivity: 55–65 W/m·K
- Vickers hardness: 22–23 GPa
- Crystal structure: layered, strong B-B covalent bonds + Zr-B metallic bonds

1.2 Raw Material Precursors

Zirconium source: - Primary mineral: **Zircon (ZrSiO₄)** — found in heavy mineral beach sands (Australia, South Africa, India, USA) - Major producing countries: Australia (#1, ~35% world supply), South Africa (#2), China (#3) - Zircon to ZrO₂: Chlorination or fusion-alkali processing removes silicon → ZrO₂ - ZrO₂ to Zr metal: Kroll process (reduction with Mg) → then boronization OR direct synthesis

Boron source: - Primary mineral: **Borax (Na₂B₄O₇·10H₂O)** — Boron, California (US Borax/Rio Tinto) and Turkey (#1 global) - Also: **Colemanite (Ca₂B₆O₁₁·5H₂O)**, **Kernite** — all from evaporite deposits - Borax to B₂O₃ to B metal: electrochemical reduction or carbothermal reduction at 1500°C

ZrB₂ synthesis routes: 1. **Boro-carbothermal reduction (industrial standard):** ZrO₂ + B₂O₃ + C → ZrB₂ + CO₂ (at 1600–1800°C in vacuum/inert atmosphere) — Most common commercial route; batch process in induction furnaces 2. **Direct synthesis from elements:** Zr powder + 2B → ZrB₂ (SHS — Self-propagating High-temperature Synthesis) — Faster, less equipment, suitable for in-house at pilot scale 3. **Sol-gel precursor route (2025 advance):** Zirconium n-propoxide + boric acid in isopropanol → pyrolysis at 1200°C in H₂ — Produces submicron powder (<80 nm) at lower temp; lower cost than commercial powder 4. **Plasma synthesis (CO₂ route, NIMS Japan 2025):** CO₂ plasma + ZrCl₄ + BCl₃ → ZrB₂ + CO + HCl — Carbon-neutral synthesis using captured CO₂; emerging industrial relevance

1.3 Commercial Acquisition

Supplier	Country	Grade	Price/kg (2026)	MOQ
American Elements	USA	99.5%, <10 μm	\$380–\$420	100 g
Alfa Aesar (Thermo Fisher)	USA	98%, -325 mesh	\$340–\$380	250 g

Supplier	Country	Grade	Price/kg (2026)	MOQ
Treibacher Industrie AG	Austria	99%+, 2-5 μm	\$280–\$320	1 kg
Höganäs AB	Sweden	Industrial, 5 μm	\$220–\$260	5 kg
BGRIMM Technology	China	98%, various	\$120–\$160	1 kg
H.C. Starck (Materion)	Germany/USA	99.5%, 1-3 μm	\$400–\$450	500 g

Volume pricing: At 1,000 kg/yr: \$180–\$240/kg (domestic). At 10,000 kg/yr: \$130–\$180/kg

1.4 In-House Synthesis — Feasibility Assessment

SHS (Self-propagating High-temperature Synthesis) — RECOMMENDED IN-HOUSE ROUTE:

Equipment needed: - Inert-atmosphere reaction vessel (stainless, argon-purged): \$8,000–\$15,000 - Ball mill for powder mixing: \$2,000–\$5,000 - Sieve and particle size analysis: \$3,000 - Safety enclosure (fines/fire hazard): \$5,000

Process: 1. Mix Zr powder (99%, -100 mesh) + crystalline B (95%, -325 mesh) in 1:2 molar ratio 2. Cold-press into pellets at 100 MPa 3. Ignite in argon atmosphere — reaction is self-propagating (exothermic) 4. Mill product to target particle size 5. Quality check: XRD phase purity, laser diffraction particle size

In-house cost estimate: - Zr powder: \$45/kg (industrial grade commercial), B powder: \$30/kg - Reagent cost per kg ZrB₂: ~\$58–\$72/kg - Labor and energy: \$35/kg - **Total in-house: ~\$95–\$110/kg vs. \$220–\$420 commercial - Savings: 50–75% compared to buying finished powder**

Feasibility Rating: 4/5 — SHS is well-documented, low equipment cost, significant savings. Main risk: SHS is exothermic/ignition-sensitive — requires basic safety protocols.

1.5 Recycled / Secondary Material Pathways

- **From Phoenix Protocol zirconia recovery:** Used Aegis-C panels $\square$ H₂SO₄ leach $\square$ ZrO₂ precipitate $\square$ re-boronization Recovery efficiency: 85–92% Zr; adds \$20/kg re-processing cost
- **Industrial zirconia waste:** Zirconium-containing slags from nuclear fuel processing, ceramics grinding swarf — Available from UNF (used nuclear fuel) facility waste streams; requires qualification

CSMFAB Products Using ZrB₂: 001 (Aegis-C panel), 002 (stove burner), 006 (water heater), 009 (rebar coating), 010 (fasteners), 019 (forklift), 024 (footwear heel), 031 (seat rail), 038 (haptic insole), CSMIC 105 (power line)

Material 2 — Silicon Nitride (Si₃N₄)

Formula: Si₃N₄ | **Crystal:** alpha (amorphous/metastable) and beta (stable, hexagonal) | **Class:** Engineering ceramic

2.1 Physical Properties

- Melting/decomposition: 1900°C (decomposes; no true melt)
- Density: 3.17–3.21 g/cm³
- Fracture toughness (with sintering aids): 5–8 MPa·m^{0.5}
- Flexural strength: 700–1200 MPa (sintered grades)
- Thermal shock resistance: excellent (low alpha, high k)

2.2 Raw Material Precursors and Synthesis

Silicon source: - **Quartzite (SiO₂)** — mined in Virginia, Arkansas, Norway, Brazil - Silica to metallurgical silicon: electric arc furnace reduction with carbon - Si to Si₃N₄: two primary routes:

Route A — Direct Nitridation (lowest cost, in-house viable): Si(s) + N₂(g) → Si₃N₄ (at 1300-1400°C in N₂ atmosphere) - Uses silicon metal powder (\$2.50/kg) and nitrogen gas (\$0.08/m³ industrial) - Reaction exothermic; requires temperature control - Produces alpha-Si₃N₄; needs sintering aids (Y₂O₃, Al₂O₃) for densification

Route B — Carbothermal Nitridation: 3SiO₂ + 6C + 2N₂ → Si₃N₄ + 6CO (at 1500°C) - Cheaper silica feedstock vs. Si metal

Route C — Chemical Vapor Deposition (for films; not bulk): SiCl₄ + NH₃ → Si₃N₄ + HCl

2.3 Commercial Acquisition

Supplier	Price/kg (2026)	Grade
Ube Industries	\$28-35/kg	UBE SN-E10 (sinterable)
H.C. Starck (Materion)	\$32-40/kg	LC12S (fine, <1 μm)
Shanghai Greenearth	\$18-25/kg	Industrial grade
Ceradyne (3M)	\$35-45/kg	Aerospace grade
Stanford Advanced Materials	\$22-30/kg	Research grade

2.4 In-House Synthesis

Direct nitridation — viable at pilot scale: Equipment: Tube furnace (1500°C, N₂ atmosphere): \$12,000–\$25,000 Raw materials: metallurgical Si powder \$2.50/kg, N₂ gas \$0.08/m³ In-house cost: **\$8–\$14/kg** (vs. \$18–\$45 commercial) **Feasibility Rating: 4/5** — straightforward chemistry; N₂ atmosphere standard in ceramics labs.

2.5 Sintering Aids (Co-materials)

Si₃N₄ densification requires: - **Yttria (Y₂O₃):** 2–8 wt% — see Critical Minerals (Part V) - **Alumina (Al₂O₃):** 0–4 wt% — \$2.50/kg technical grade (abundant) - These form liquid phase at sintering temp → dense Y-Al-Si-N oxynitride glass phase

CSMFAB Products Using Si₃N₄: 002 (stove burner base), 007 (oven components), CSMMECH 001 (robot gears + bearings), 019 (forklift bearing housings), 020 (motor rotor), 031 (seat rail)

Material 3 — Silicon Carbide (SiC)

Formula: SiC | **Crystal:** Cubic (3C-beta) or hexagonal (6H, 4H-alpha) | **Class:** Engineering ceramic

3.1 Physical Properties

- Melting/decomposition: 2730°C (sublimation)
- Density: 3.21 g/cm³ (sintered dense)
- Hardness: 26–28 GPa Vickers
- Thermal conductivity: 120 W/m·K (single crystal); 25-50 W/m·K (polycrystalline sintered)
- Electrical resistivity: variable, 1-10 Ω·cm (undoped); >10¹² Ω·cm (high-purity undoped)

3.2 Raw Materials and Synthesis

Acheson Process (dominant industrial method since 1893): $\text{SiO}_2 + 3\text{C} \rightarrow \text{SiC} + 2\text{CO}$ (at 2200–2500°C in graphite resistance furnace) - Raw materials: silica sand (\$0.08/kg) + petroleum coke (\$0.25/kg) - **The lowest cost advanced ceramic synthesis route in existence** - Large-scale production: Saint-Gobain, ESK/3M, Fiven, ELK (Russia)

Advanced powder synthesis for fine particles (for Aegis): - Chemical vapor deposition (CVD): $\text{SiH}_4 + \text{CH}_4 \rightarrow \text{beta-SiC}$ (expensive; for fibers/coatings) - Carbothermal: $\text{SiO}_2 + \text{C}$ precursor (sol-gel route) $\rightarrow$ submicron SiC

3.3 Commercial Acquisition

Supplier	Grade	Price/kg (2026)	Notes
Saint-Gobain (Hexoloy)	F800 sinterable	\$18–\$25/kg	Dominant US supplier Norwegian; volume pricing
Fiven (formerly Elkem)	SiC GC	\$12–\$18/kg	
ESK-SiC (3M)	Various	\$20–\$30/kg	Sintering-grade Abrasive grade — not sinterable
CR International	Crude SiC	\$2–\$5/kg	
Roper Technologies	Fibers (Nicalon)	\$800–\$2,000/kg	Continuous fiber for composites
COI Ceramics	CVDS SiC	\$1,200–\$2,500/kg	Defense/aerospace grade

Beta-SiC nanofibers (for toughening): \$200–\$600/kg depending on purity

3.4 In-House Synthesis

Modified Acheson in small electric furnace: Equipment: Graphite resistance heating furnace, 1800°C: \$18,000–\$35,000 Raw materials: Quartz sand \$0.08/kg, petroleum coke \$0.25/kg In-house cost: **\$4–\$8/kg** (crude SiC); add \$12–\$20/kg for milling to sinterable grade **Feasibility Rating:** 3/5 — energy intensive; viable for mid-scale operation with local power

3.5 Applications in CSMFAB

SiC is used primarily as: - Reinforcing phase in ZrB₂-SiC composite (15-20 vol.% whiskers or particles) - Infrared heating elements in stoves and ovens (electrical-grade SiC rods) - Continuous fiber reinforcement in high-temp composites

Products: 001, 002, 003 (ZrB₂-SiC structural), 007 (SiC heating rod), 008 (SiC element), CSMIC 107

Material 4 — Yttria-Stabilized Zirconia (3Y-TZP; ZrO₂)

Formula: ZrO₂ + 3 mol% Y₂O₃ | **Class:** Transformation-toughened ceramic

4.1 Physical Properties

- Melting point: 2715°C (ZrO₂ pure)
- Density: 6.05 g/cm³ (3Y-TZP)
- Fracture toughness: 8–12 MPa·m^{0.5} (best of oxide ceramics)
- Flexural strength: 1200–1500 MPa (dental-grade)
- Hardness: 12–14 GPa

- Electrical resistivity: $>10^{12} \Omega\cdot\text{m}$ (roomtemp) — excellent GIC insulator

4.2 Raw Materials

Zirconia source: Same as ZrB_2 — Zircon sand $\rightarrow \text{ZrO}_2$ via: - Alkali fusion: $\text{ZrSiO}_4 + \text{NaOH} \rightarrow \text{Na}_2\text{ZrO}_3 + \text{Na}_2\text{SiO}_3 \rightarrow \text{ZrO}_2$ by acid leach - Chlorination: $\text{ZrSiO}_4 + 4\text{Cl}_2 \rightarrow \text{ZrCl}_4 \rightarrow \text{ZrO}_2$ pyrolysis

Yttria stabilizer source: - Yttrium from rare earth ores (see Part V) - Y_2O_3 added at 3 mol% to stabilize tetragonal phase at room temperature - This “metastable tetragonal” phase stress-transforms to monoclinic at crack tip $\rightarrow$ toughening

4.3 Commercial Acquisition

Supplier	Grade	Price/kg (2026)
Tosoh Corporation	TZ-3Y-E (dental/structural)	\$40–\$60/kg
Saint-Gobain ZirPro	Y-TZP microspheres	\$35–\$55/kg
Inframat Advanced Materials	99.9%, <100 nm	\$95–\$140/kg
Stanford Advanced Materials	3Y, sintering grade	\$45–\$65/kg
Zircoa Inc.	Industrial grade	\$30–\$45/kg

Machined parts (ZrO₂ blanks, rods): - Raw billets: \$25–\$60/kg compact - Machined to shape: add \$30–\$150/part (CNC grinding, carbide tooling)

4.4 In-House Synthesis

Co-precipitation route: $\text{ZrOCl}_2(\text{aq}) + \text{YCl}_3(\text{aq}) \rightarrow$ add $\text{NH}_3 \rightarrow \text{Zr-Y}$ hydroxide co-precipitate $\rightarrow$ calcine $\rightarrow$ 3Y-TZP powder

Equipment: pH-controlled precipitation reactor (\$8,000), filter press (\$5,000), calcination furnace (\$12,000) Raw materials: $\text{ZrOCl}_2 \sim \$12/\text{kg}$; $\text{YCl}_3 \sim \$35/\text{kg}$; NH_3 bulk \$0.40/kg In-house cost: **\$25–\$38/kg** (vs. \$35–\$140 commercial) **Feasibility Rating:** 4/5 — standard ceramic powder precipitation; well-documented.

4.5 Products and Applications

Used as: structural fasteners, door handles, hinges, grinding inserts, fuel injector tips, water heater fittings, pipeline joints. **Products:** 006, 010, 012, 015, 016, 028, 031, 034, 035, CSMIC 105, 106, CSMMECH 002

Material 5 — Alumina (Al_2O_3) and Corundum

Formula: Al_2O_3 | **Crystal:** Rhombohedral (corundum) | **Class:** Oxide ceramic

5.1 Physical Properties

- Melting point: 2072°C
- Density: 3.99 g/cm³ (single crystal); 3.80–3.95 sintered
- Hardness: 18–20 GPa
- Electrical resistivity: $>10^{14} \Omega\cdot\text{m}$

5.2 Raw Materials: The Most Accessible Ceramic

Source mineral: Bauxite ($\text{Al}_2\text{O}_3 \cdot n\text{H}_2\text{O}$) - World's most abundant ore after SiO_2 - Largest reserves: Australia (33%), Vietnam (21%), Guinea (20%) - US domestic: Arkansas, Alabama (minor production) - Bayer process: Bauxite + NaOH $\rightarrow$ $\text{Al}(\text{OH})_3$ precipitate $\rightarrow$ calcine $\rightarrow$ Al_2O_3 - Cost of Bayer Al_2O_3 : **\$2.00–\$3.50/kg** — most cost-accessible advanced ceramic

High-purity synthesis: - Alcoa, Almatris: calcined technical-grade Al_2O_3 \$3–\$8/kg - 99.99%+ purity via zone refining or Verneuil process for corundum: \$25–\$80/kg

5.3 Commercial Pricing

Grade	Price/kg (2026)	Use
Technical (99%, Bayer)	\$3–\$6/kg	Structural, electrical insulation
High-purity (99.9%, calcined)	\$8–\$15/kg	Ceramic armor, precision parts
Sapphire (synthetic single crystal)	\$500–\$2,000/kg	Windows, wear
ALON (polycrystalline transparent)	\$800–\$1,200/kg	Transparent armor

5.4 In-House Production

Producing calcined Al_2O_3 from bauxite requires Bayer-scale alkali plant — **not practical in-house**. However, **purchasing technical Al_2O_3 at \$3–\$6/kg is the lowest-cost material acquisition in the portfolio** and no in-house benefit exists at this price.

ALON synthesis (for transparent armor applications): $\text{AlN} + \text{Al}_2\text{O}_3$ at 1800°C in N_2 $\rightarrow$ ALON Equipment investment: \$45,000–\$90,000 (pressurized hot press) Cost differential: ALON commercial \$800–\$1,200/kg vs. in-house estimate \$180–\$280/kg **In-house ALON: Feasibility Rating 3/5** — viable for robotics armor at scale.

Products: Used as sintering aid in Si_3N_4 and ZrO_2 ; ceramic-coated motor windings; CSMMECH armor.

Material 6 — MAX Phase Ceramics: Ti_3AlC_2 and Ti_2AlC

Formula/Class: $\text{Mn}+1\text{AX}_n$ layered ternary carbide/nitride | Hybrid metal-ceramic

6.1 Physical Properties

- Density Ti_3AlC_2 : 4.2 g/cm^3
- Flexural strength: 250–350 MPa
- Fracture toughness: $5\text{--}12 \text{ MPa}\cdot\text{m}^{0.5}$ (kinking/nonlinear elastic — no brittle fracture)
- Machinable with carbide tools (key advantage)
- Electrical conductivity: 10^6 S/m (metallic) — isolated use only for GIC immunity
- Max service temperature: 1400°C (Ti_3AlC_2)

6.2 Raw Material Precursors

Elements required: Ti + Al + C (and/or TiC)

Titanium source: - Mineral: **Ilmenite (FeTiO_3)** or **Rutile (TiO_2)** — beach sands, Virginia, Florida, Australia - Kroll process: $\text{TiCl}_4 + \text{Mg} \rightarrow \text{Ti metal}$ (expensive; $\sim \$20/\text{kg}$) - OR use TiO_2 directly in carbothermal synthesis of MAX phase

Aluminum source: Alumina (Bayer) $\rightarrow$ smelted Al; \$2/kg bulk aluminum metal

Carbon source: Graphite powder \$2–\$8/kg; highly available

MAX Phase synthesis routes:

Route A — Solid-state reactive sintering (most practical): $3\text{TiC} + \text{Ti} + 2\text{Al} \rightarrow \text{Ti}_4\text{AlC}_3$ (or vary stoichiometry for Ti_3AlC_2) - Mix TiC, Ti, Al powders $\rightarrow$ press $\rightarrow$ heat to 1400°C in argon $\rightarrow$ MAX phase forms - Equipment: regular box furnace + hydraulic press - Time: 2 hours at temperature

Route B — Titanium + aluminum + graphite: $3\text{Ti} + 2\text{Al} + 2\text{C} \rightarrow \text{Ti}_3\text{AlC}_2$ (carbothermal, $1300\text{--}1500^\circ\text{C}$ in argon)

Route C — Hot pressing in argon: highest density, 500 MPa + 1400°C

6.3 Commercial Acquisition

Supplier	Price/kg (2026)	Format
Maxthal (Kanthal/Sandvik)	\$320–\$420/kg	Powder, sintered tile
3-ONE-2 (USA startup)	\$280–\$380/kg	Ti_3AlC_2 , -325 mesh
Laizhou Kai Kai Ceramic	\$80–\$120/kg	Chinese industrial grade
Sigma-Aldrich / Merck	\$600–\$900/kg	Research grade

Production scale: MAX phase is still relatively new to industrial production; prices will fall. 2028-2029 projection: \$150–\$250/kg as Sandvik/Maxthal scales up.

6.4 In-House Synthesis — STRONGLY RECOMMENDED

Equipment: hot press or pressureless sintering furnace (1500°C , inert) already needed for ZrB_2 . Raw material cost: TiC \$8/kg + Ti \$20/kg + Al \$2/kg $\rightarrow$ Ti_3AlC_2 yield **In-house cost estimate: \$35–\$55/kg** (vs. \$80–\$420 commercial) **Feasibility Rating: 5/5** — uses same equipment as Si_3N_4 synthesis; shared infrastructure.

Products: 001, 010, 019, CSMMECH 001 (as machinable surface layer)

Material 7 — Chemically Bonded Phosphate Ceramic (CBPC) / Geopolymer

Class: Inorganic polymer; cold-set ceramic; non-oxide cement-analogue

7.1 Composition (CBPC — Magnesium Phosphate)

- MgO (dead-burned magnesia) + KH_2PO_4 (potassium dihydrogen phosphate) + water
- Sets at room temperature in 15–30 minutes; no kiln required
- Compressive strength: 30–80 MPa; rises to 100+ with aggregates

Geopolymer (alkali-activated): - Fly ash (class F) + NaOH (8M) + $\text{Na}_2\text{SiO}_3 \rightarrow$ geopolymer gel $\rightarrow$ hardens - 78% lower CO_2 vs. Portland cement; 80 MPa at 28 days

7.2 Raw Material Sources

CBPC Materials: - MgO: magnesite (MgCO_3) mining $\rightarrow$ calcination $\rightarrow$ magnesia; Turkey, China, Russia. \$0.35–\$0.80/kg - KH_2PO_4 : phosphate rock mining (Florida, Morocco) $\rightarrow$ H_3PO_4 + KOH neutralization. \$1.20–\$2.50/kg

Geopolymer Materials: - Fly ash: Class F — sourced FREE or at minimal cost from coal power plants (waste product) - Metakaolin: calcined kaolin clay (\$0.55/kg); abundant worldwide - GGBS: ground granulated blast furnace slag — steel mill byproduct (\$0.18/kg) - NaOH: Chlor-alkali process; globally available commodity (\$0.45/kg 8M solution) - Na₂SiO₃: silica + NaOH or by-product of silica gel production (\$0.55/kg)

7.3 In-House Mixing — TRIVIAL

No kiln, no special equipment for geopolymer: Mixer + molds → pour → ambient cure. Equipment cost: \$2,000–\$8,000 (concrete-style mixing station). In-house cost per m³: **\$280–\$380** (vs. Portland concrete \$150–\$250/m³ baseline; quality justified)

CBPC in-house: Same mixer; add phosphate addition and temperature control. In-house cost: **\$85–\$120/m³** for CBPC without reinforcement

Feasibility Rating: 5/5 — simplest in-house processing of any CSMFAB material.

Products: 006 (CBPC water heater shell), CSMIC 107 (geopolymer concrete), CSMMECH 003 (structural foundation)

Material 8 — Silicon Carbide Heating Elements

Specialty application: Resistive infrared heating rods for ovens and stoves

8.1 Construction and Specifications

- Material: Dense, electrically conducting SiC (alpha phase, metallic silicon-bonded or recrystallized)
- Operating temperature: 1400°C continuous, 1600°C peak
- Resistivity: 0.1–2.0 Ω·cm (doped with nitrogen for conductivity)
- Peak infrared emission: 890 nm (matches 10.8 W/m² FIR output per cm²)

8.2 Acquisition

Commercial suppliers: - Global (Kanthal): \$35–\$80/element depending on geometry - Solarc SiC (USA): \$45–\$90/element - I Squared R Elements (Starbar): \$40–\$85/element - Chinese manufacturers: \$8–\$20/element (variable quality)

In-house production: Producing SiC heating elements requires controlled resistivity sintering (nitrogen-atmosphere sintering) — requires \$80,000+ kiln with atmosphere control. **Not recommended for in-house at first.** Acquisition strategy: Purchase from Kanthal/I Squared R; negotiate volume pricing with 12-month forecast. **Feasibility Rating:** 2/5 for in-house; buy commercial.

Products: 002, 007, 008

Summary Table — Part I Ceramics

Material	2026 Commercial (/kg) In — HouseCost(/kg)	Savings	In-House Rating	
ZrB ₂ powder	\$120–\$420	\$95–\$110	50–75%	4/5
Si ₃ N ₄ powder	\$18–\$45	\$8–\$14	50–70%	4/5
SiC powder	\$12–\$30	\$4–\$8	50–75%	3/5
ZrO ₂ (3Y-TZP)	\$35–\$140	\$25–\$38	25–70%	4/5

Material	2026 Commercial (/kg) In — HouseCost(/kg)	Savings	In-House Rating	
Al2O3 technical	\$3–\$6	N/A — buy	0%	1/5
ALON	\$800–\$1,200	\$180–\$280	75–80%	3/5
Ti3AlC2 (MAX Phase)	\$80–\$420	\$35–\$55	45–85%	5/5
CBPC / Geopolymer	\$85–\$380/m ³	\$85–\$120/m ³	0–30%	5/5
SiC heating elements	\$35–\$90/ea	N/A — buy	—	2/5

Priority in-house synthesis programs (by ROI): 1. ZrB2 via SHS (50-75% savings, portfolio-wide impact) 2. Ti3AlC2 MAX Phase (45-85% savings, shared equipment with ZrB2 program) 3. Si3N4 via direct nitridation (50-70% savings) 4. 3Y-TZP via co-precipitation (25-70% savings)

End of CSMFAB Materials Study Part I | Ultra-High Temperature Ceramics Carrington Storm Motors | June 2026